

Computer Networks

Lecture 6: Switching

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- **Packet Switching**

Introduction

A network is a set of connected devices. Whenever we have multiple devices, we have the problem of how to connect them to make one-to-one communication possible.

Network Connectivity Approaches

Multiple Devices in a Network

(a) **Mesh Topology**
(Point-to-Point)



- Many direct Links
- High cost
- Most links idle
- Non scalable

(b) **Star Topology**
(Central Device)

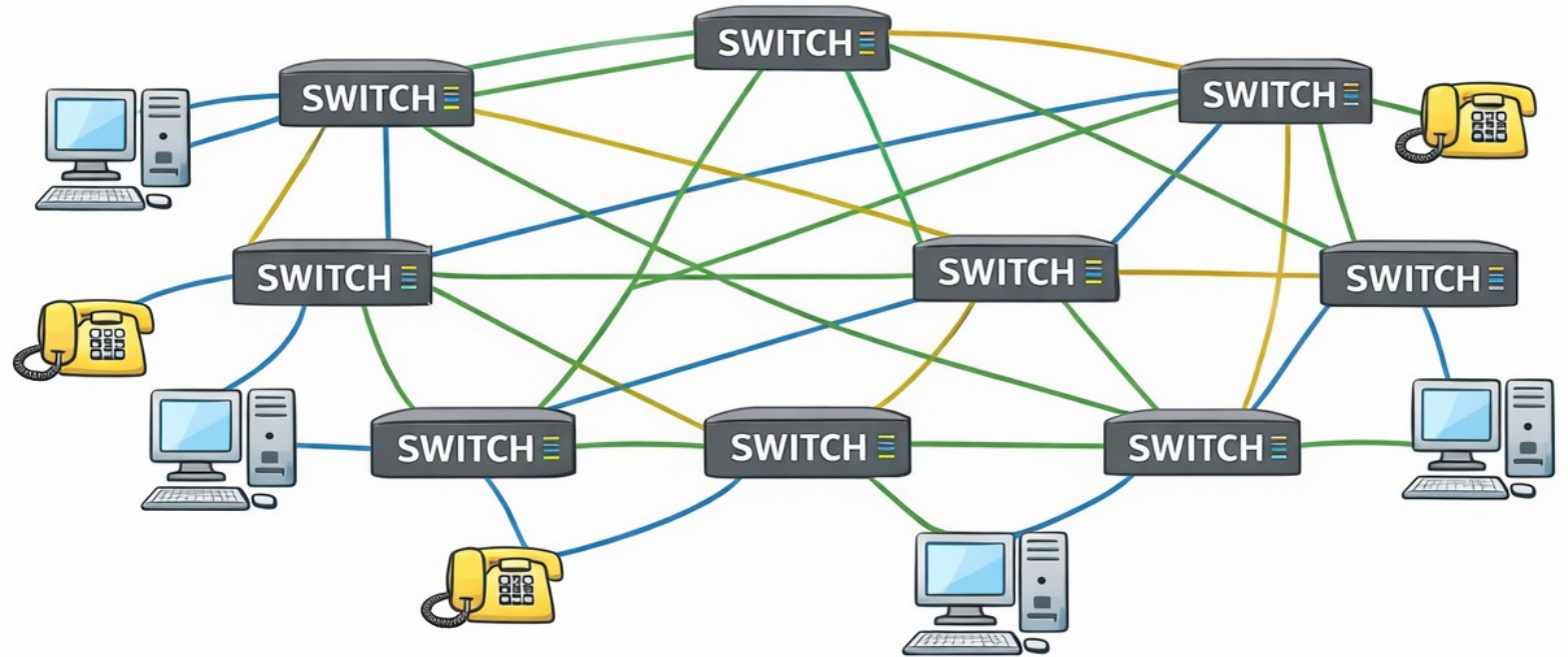


- Single central device
- Central Bottleneck
- Not Scalable

Switching

A better solution is switching.

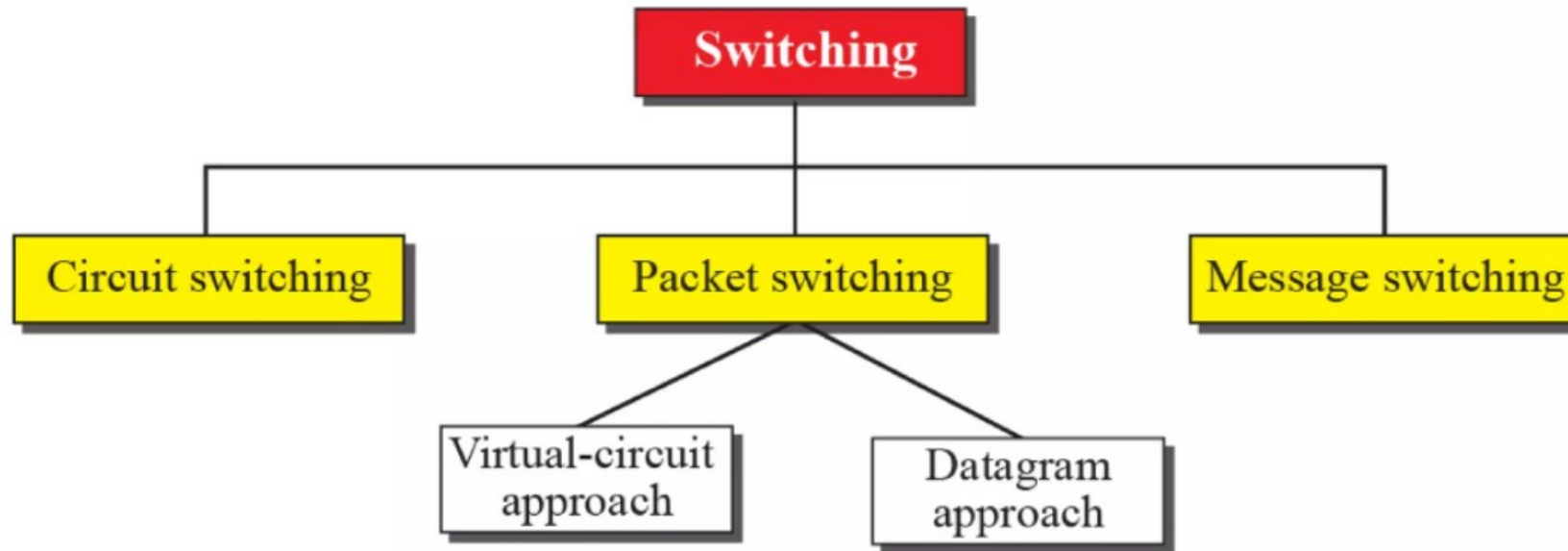
A switched network consists of a series of interlinked nodes, called switches.



Switches will create temporary connection between end systems.

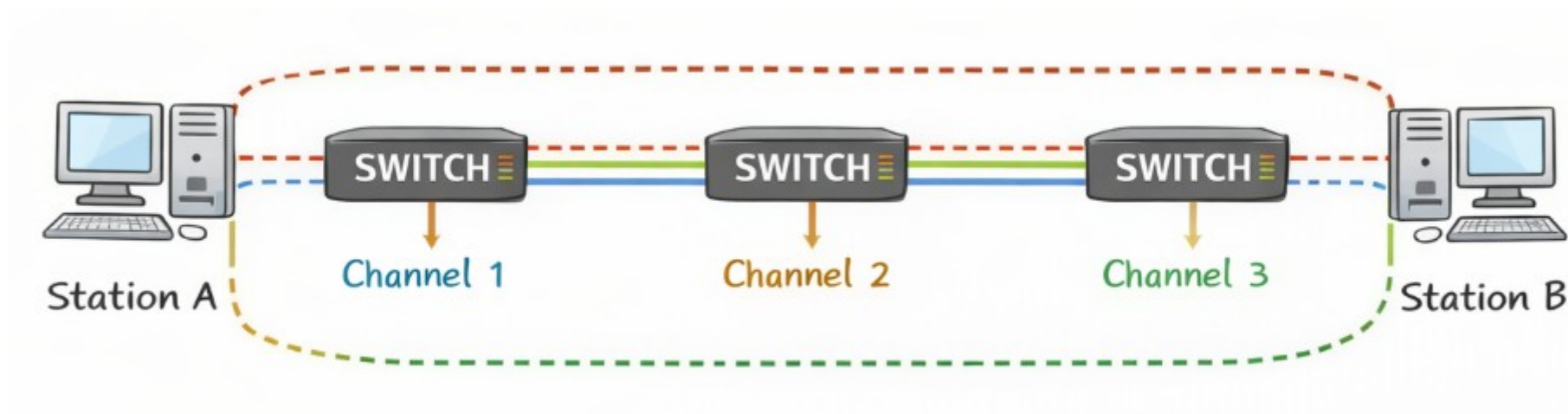
Switching Methods

Three methods of switching have been important



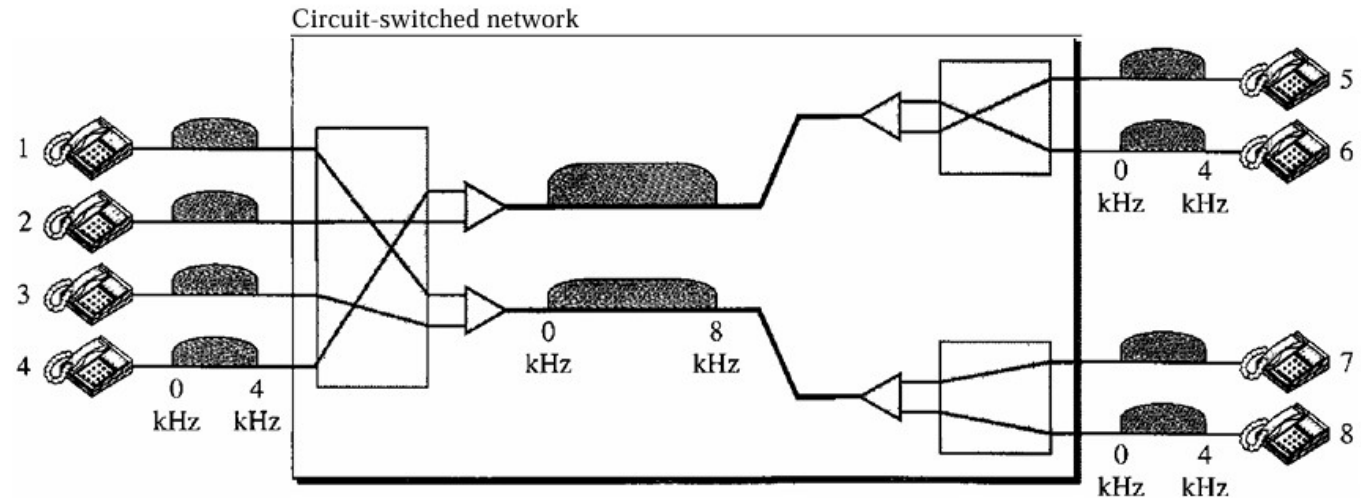
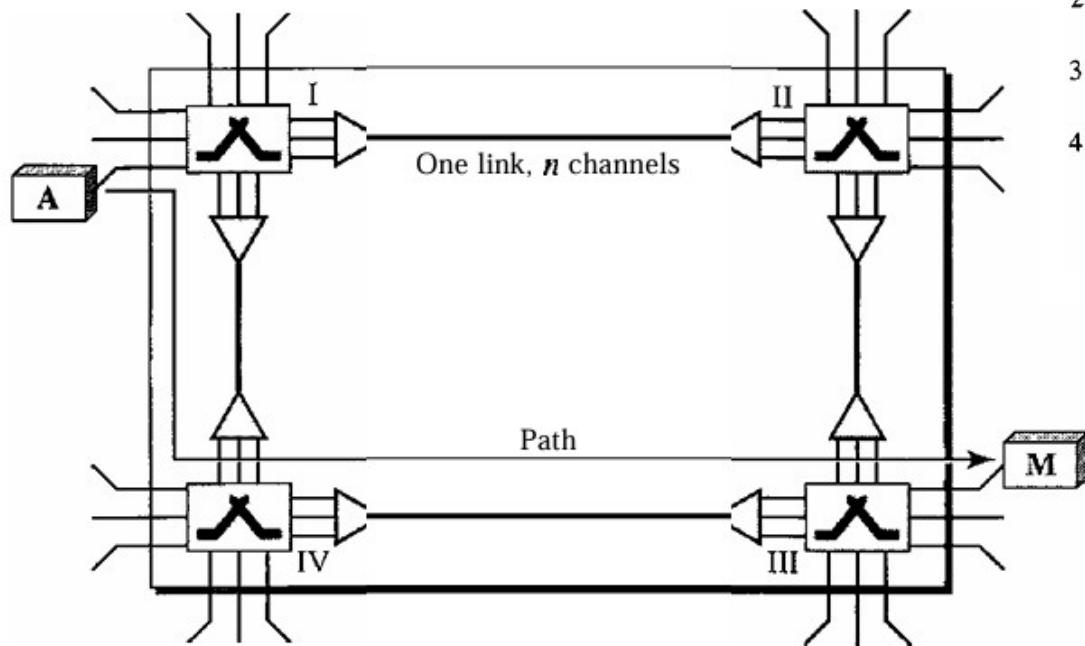
Circuit Switched Network

A circuit-switched network is made of a set of switches connected by physical links, in which each link is divided into n channels.



A dedicated path with single channel on each link uses FDM or TDM to divide the link into channel

Circuit Switched Network: Example

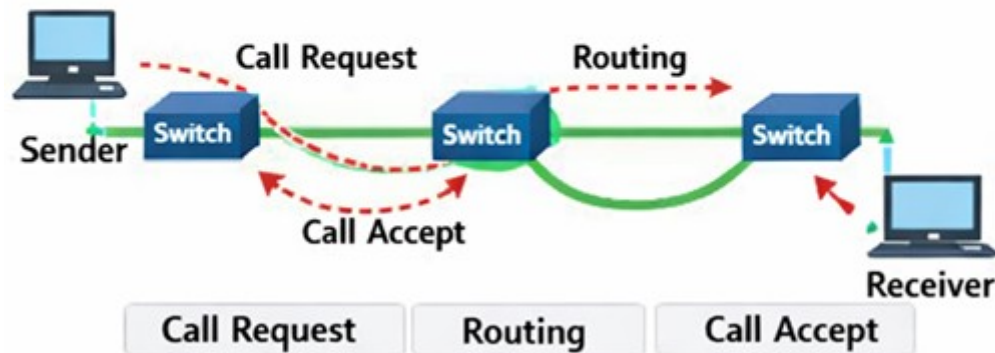


Circuit Switching: Three Phases

The actual communication in a circuit-switched network requires three phases: connection setup, data transfer, and connection teardown

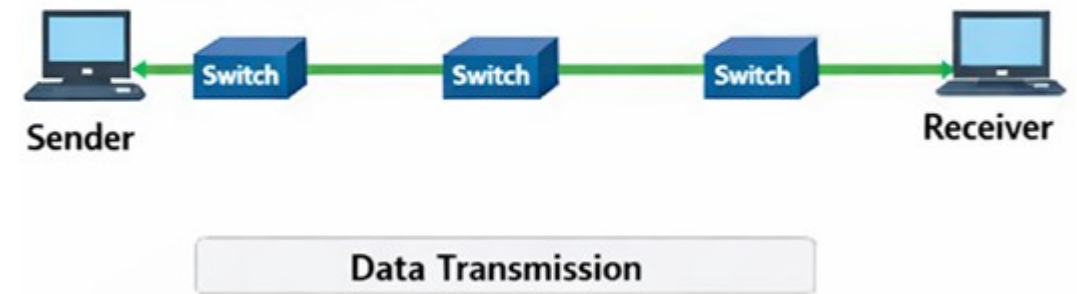
Connection Setup Phase

Establishing a dedicated path



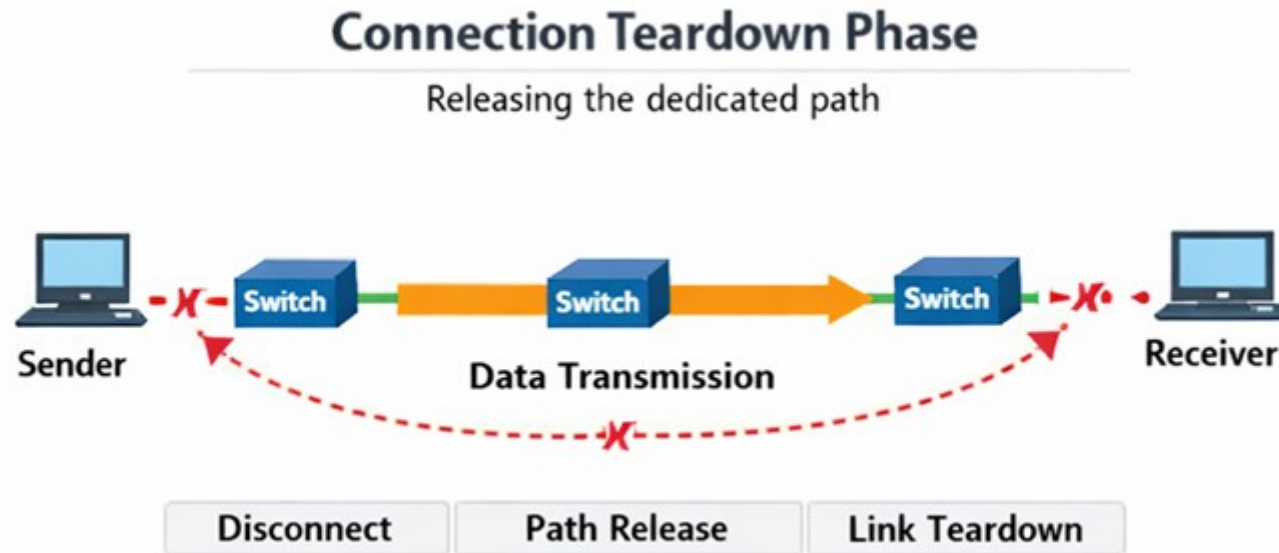
Data Transfer Phase

Transmitting the information



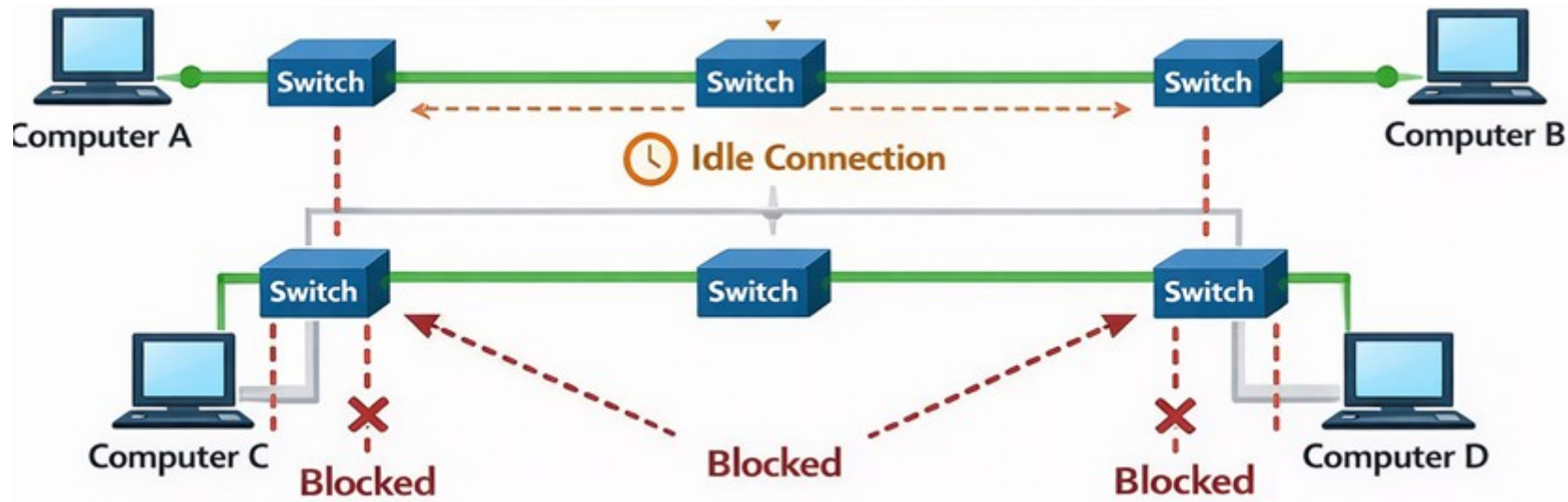
Circuit Switching: Three Phases

The actual communication in a circuit-switched network requires three phases: connection setup, data transfer, and connection teardown



Circuit Switching: Efficiency

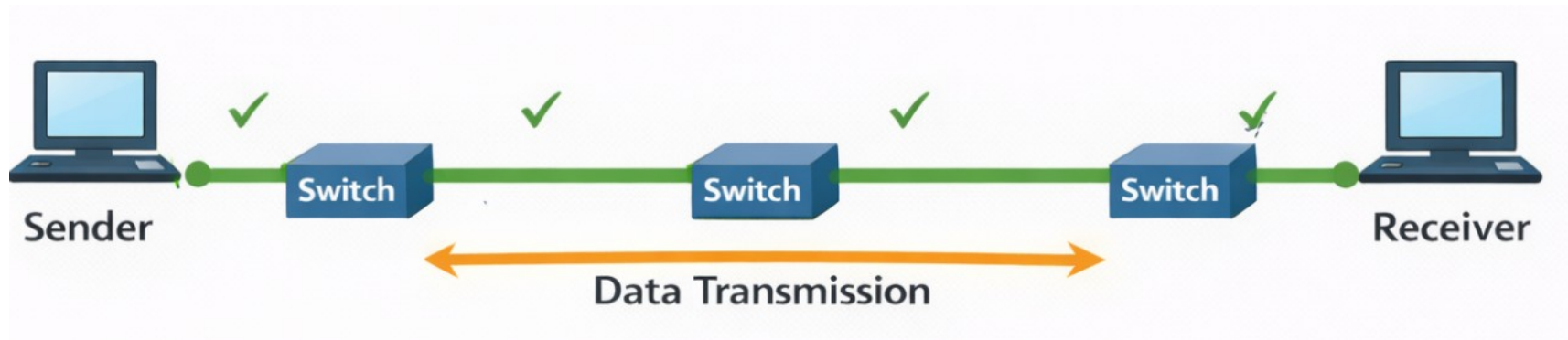
In circuit switched network, resources are reserved for the entire duration of the communication. These reserved resources are unavailable for other systems



Other systems must wait until the dedicated connection is released

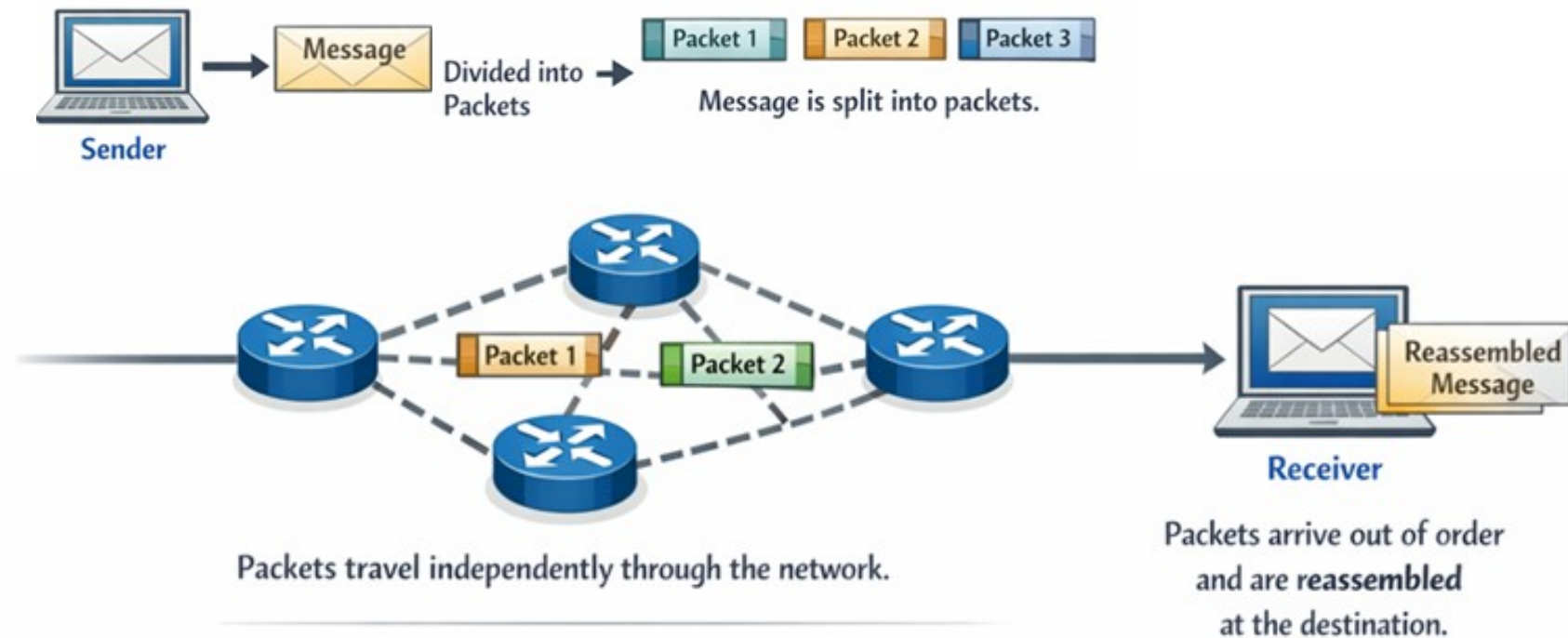
Circuit Switching: Delay

Although a circuit-switched network normally has low efficiency, the delay in this type of network is minimal. During data transfer the data are not delayed at each switch; the resources are allocated for the duration of the connection.



Packet Switching

If the message is going to pass through a packet-switched network, it needs to be divided into packets of fixed or variable size.

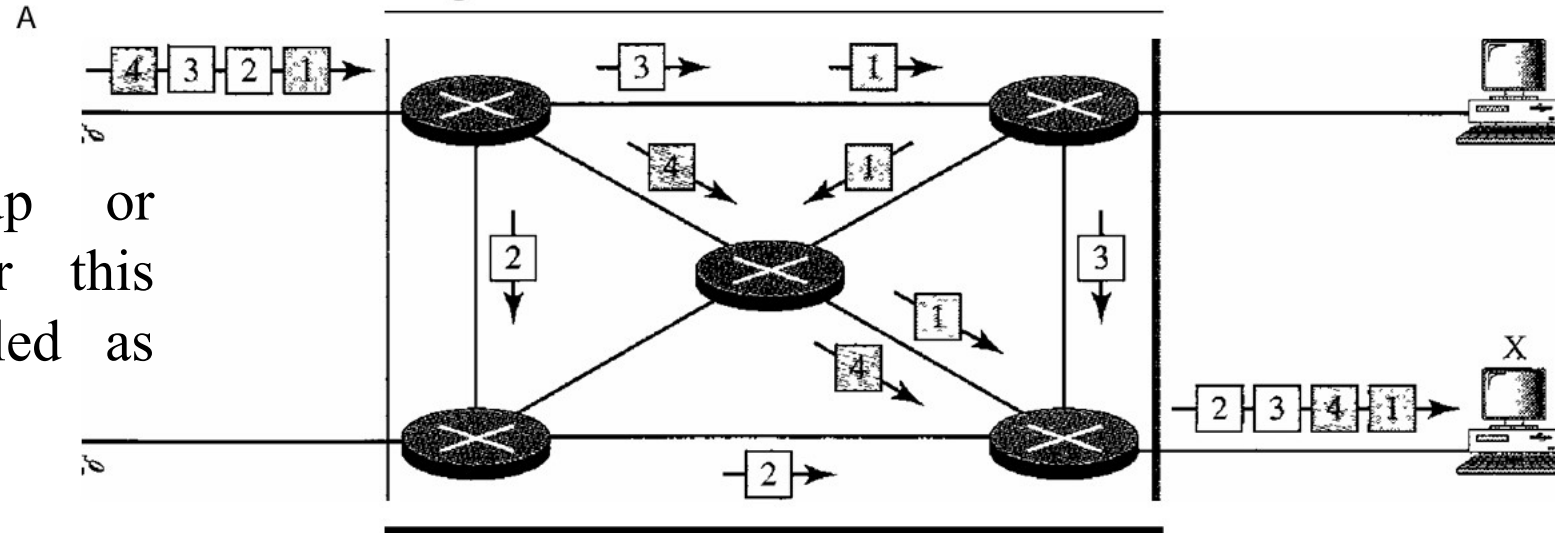


- No Reserved Bandwidth
- No Scheduled Processing Time

Datagram Network

In a datagram network, each packet is treated independently of all others. Even if a packet is part of a multipacket transmission, the network treats it as an independent entity.

Datagram network



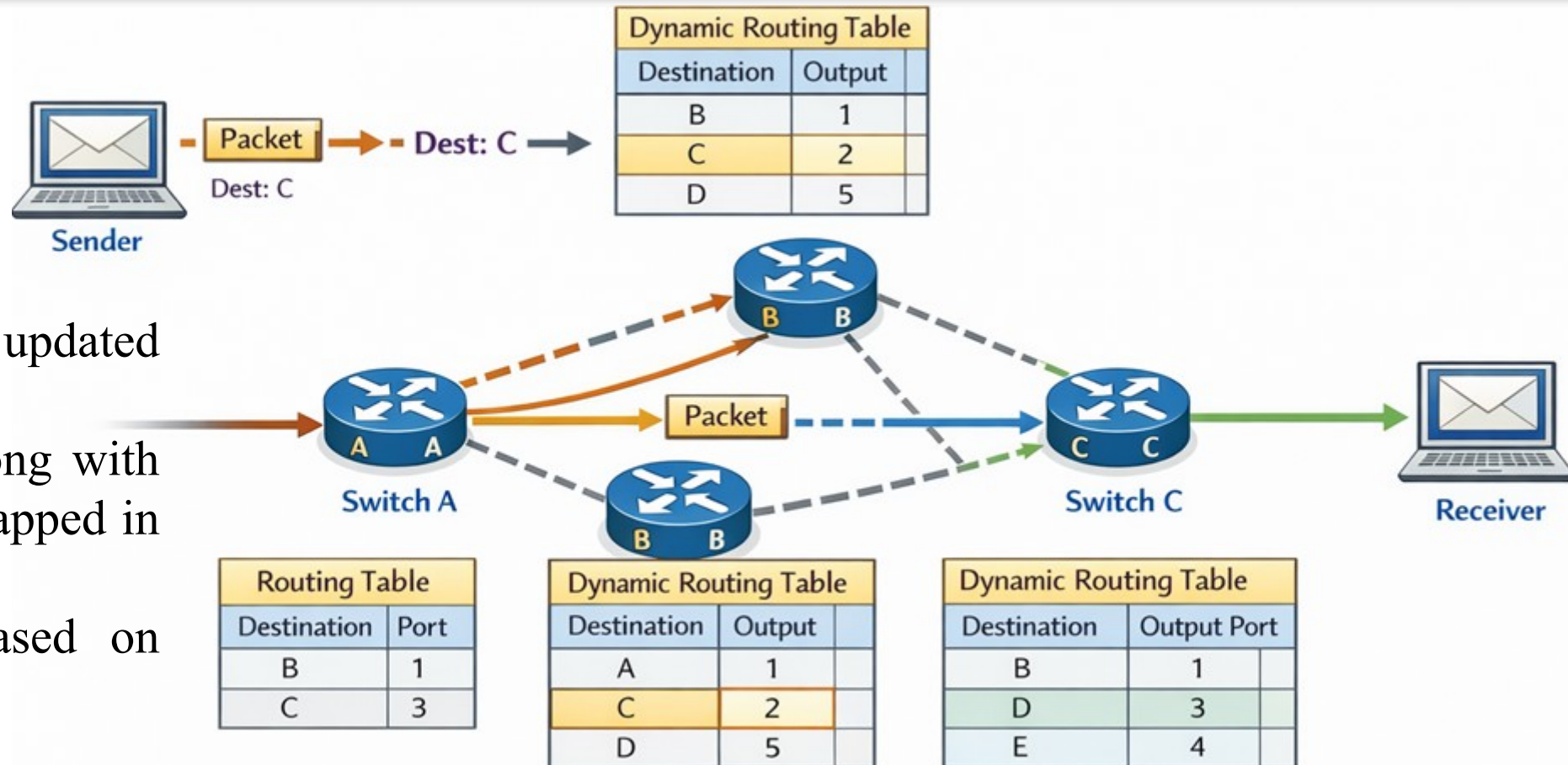
There are no setup or teardown phases. For this reason it is also called as Connectionless

Datagram switching is normally done at the network layer.

Datagram Network: Routing

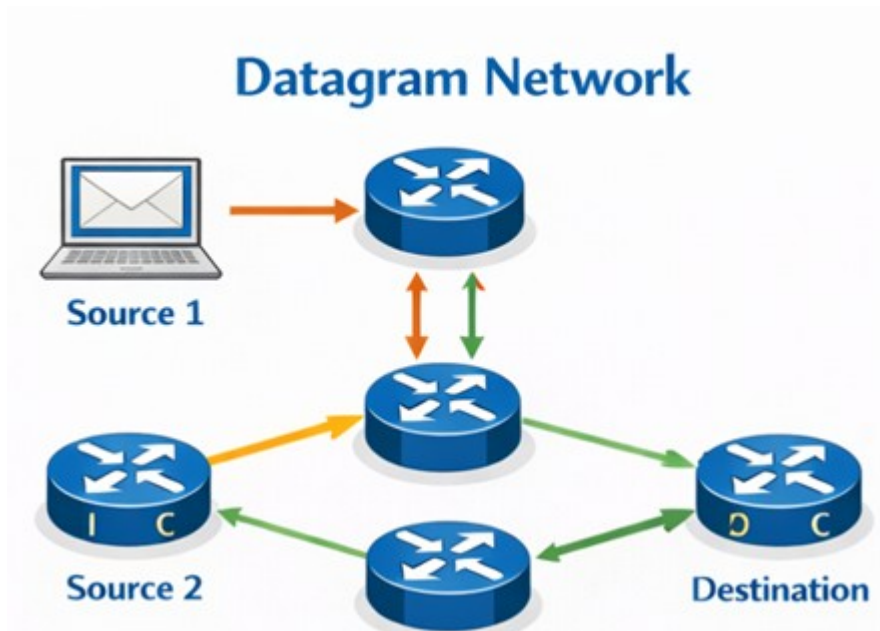
In a datagram network, each switch (or packet switch) has a routing table which is based on the destination address.

- Routing tables are updated periodically
- Destination address along with forwarding ports are mapped in routing table
- Packets are routed based on destination address

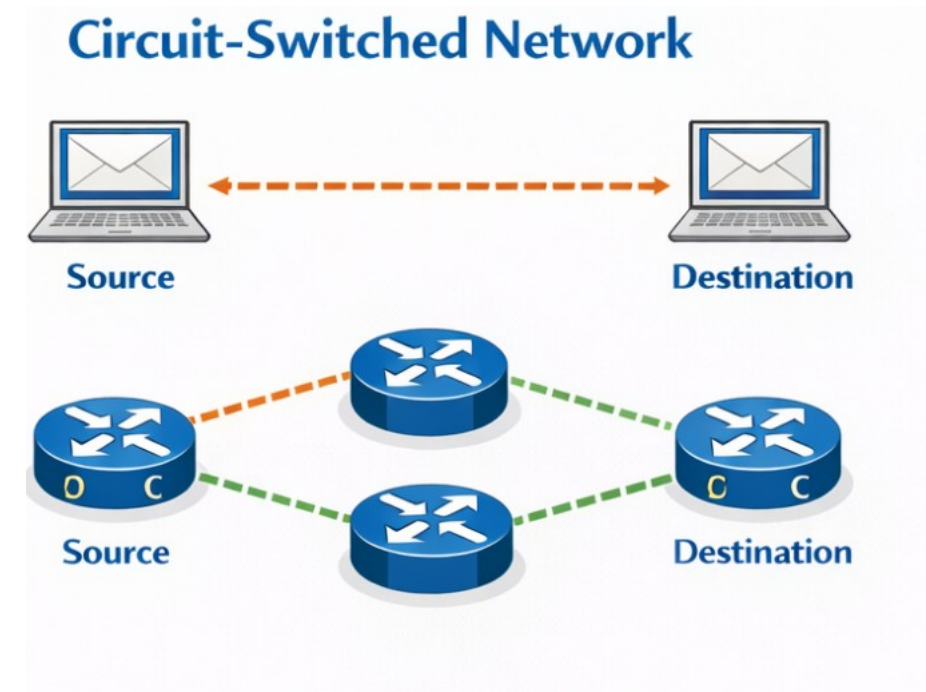


Datagram Network: Efficiency

The efficiency of a datagram network is better than that of a circuit-switched network;

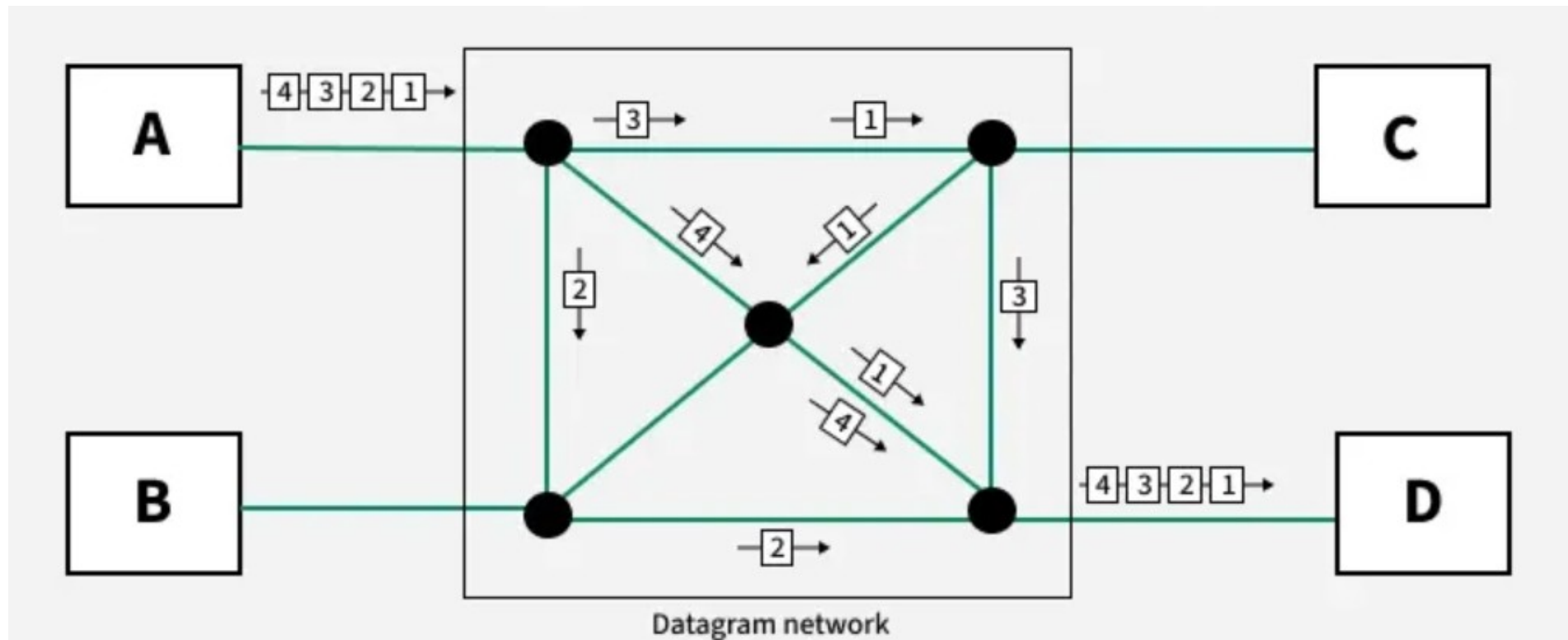


Resources are allocated only when there are packets to be transferred



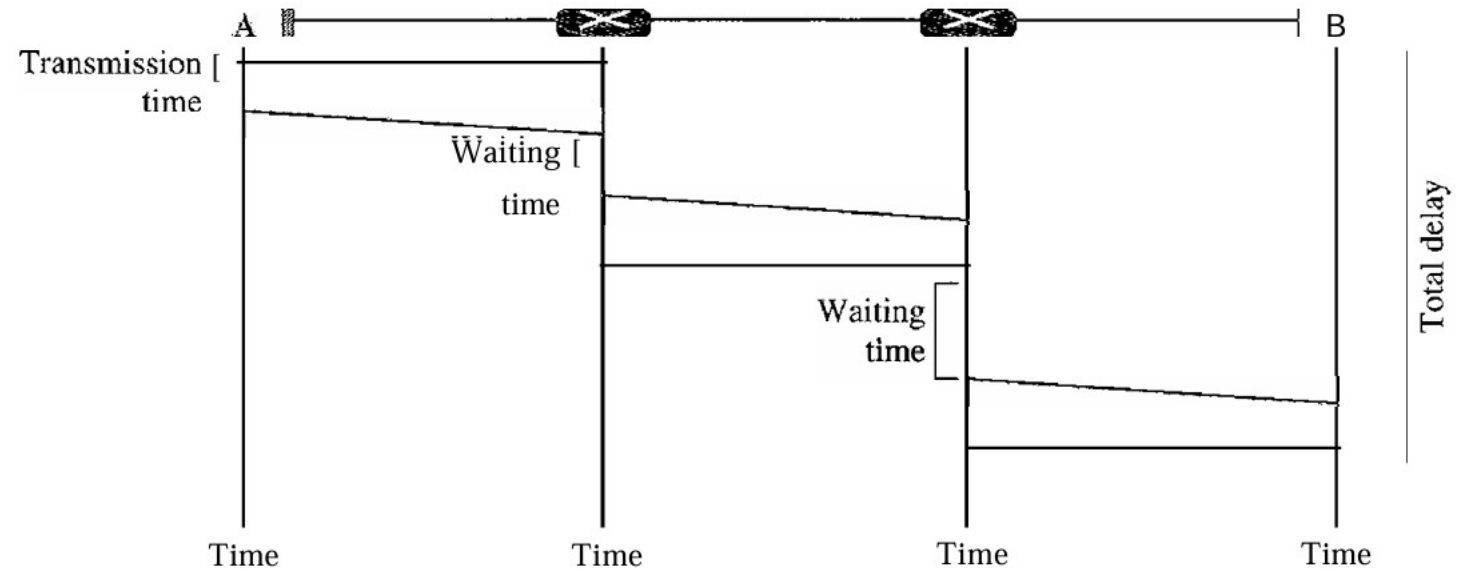
Datagram Network : Delay

Although there are no setup and teardown phases, each packet may experience a wait at a switch before it is forwarded. In addition, since not all packets in a message necessarily travel through the same switches, the delay is not uniform for the packets of a message.



Question

Consider that the data is being transmitted from source A to destination B . What is the delay in terms of transmission, propagation, Queueing and processing time?



$$\text{Total delay} = 3T + 3\tau + W_1 + W_2$$

Virtual Circuit Network

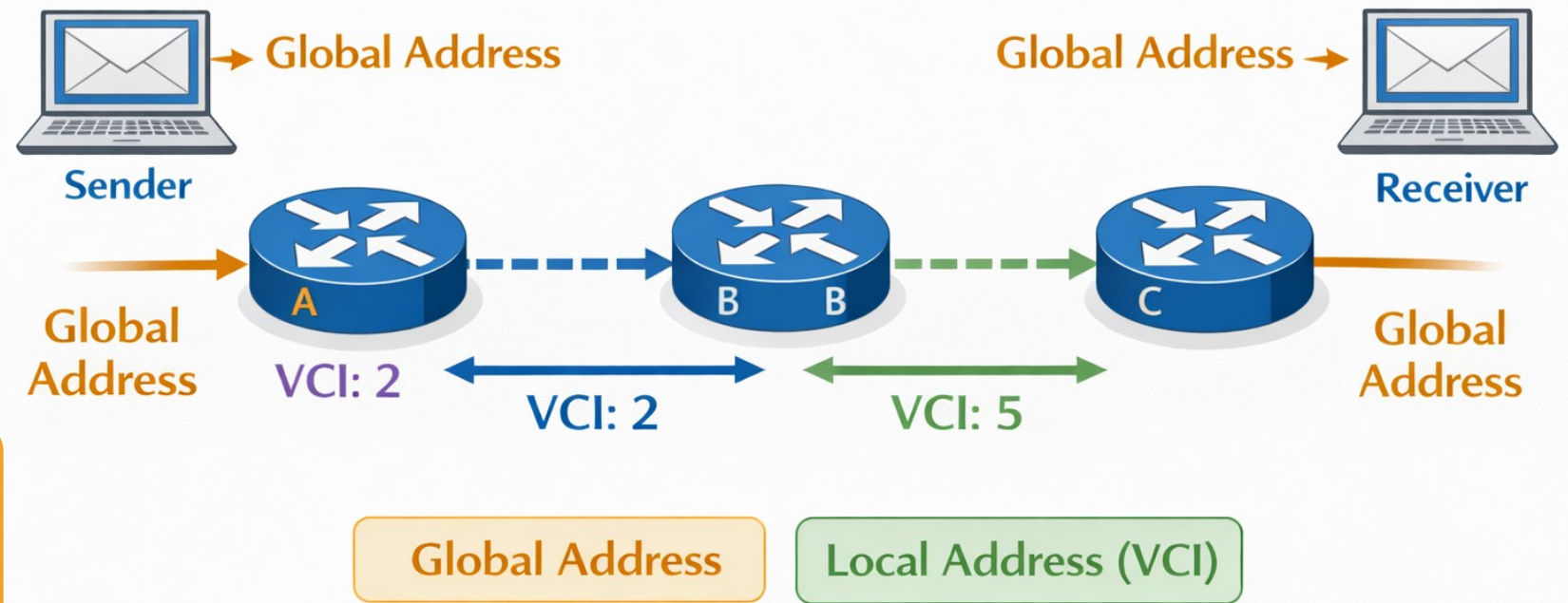
A virtual-circuit network is a combination of circuit-switched network and a datagram network. It has some characteristics of both.

- Like a circuit-switched network, a virtual-circuit network has three phases: setup, data transfer, and teardown.
- Resources may be reserved in advance during setup (like circuit switching) or allocated only when needed (like datagram networks).
- Data is divided into packets, and each packet carries an address for the next switch only, not the final destination.
- Once the connection is established, all packets travel through the same predefined path, similar to a circuit-switched network.

Virtual Circuit Network: Addressing

In a virtual-circuit network, two types of addressing are involved: global and local.

A source or a destination needs to have a global address—an address that can be unique in the scope of the network



The identifier that is actually used for data transfer is called the virtual-circuit identifier (VCI).

Virtual Circuit Network: 3-Phases

A virtual-circuit network has three phases: setup, data transfer, and teardown, similar to a circuit-switched network.

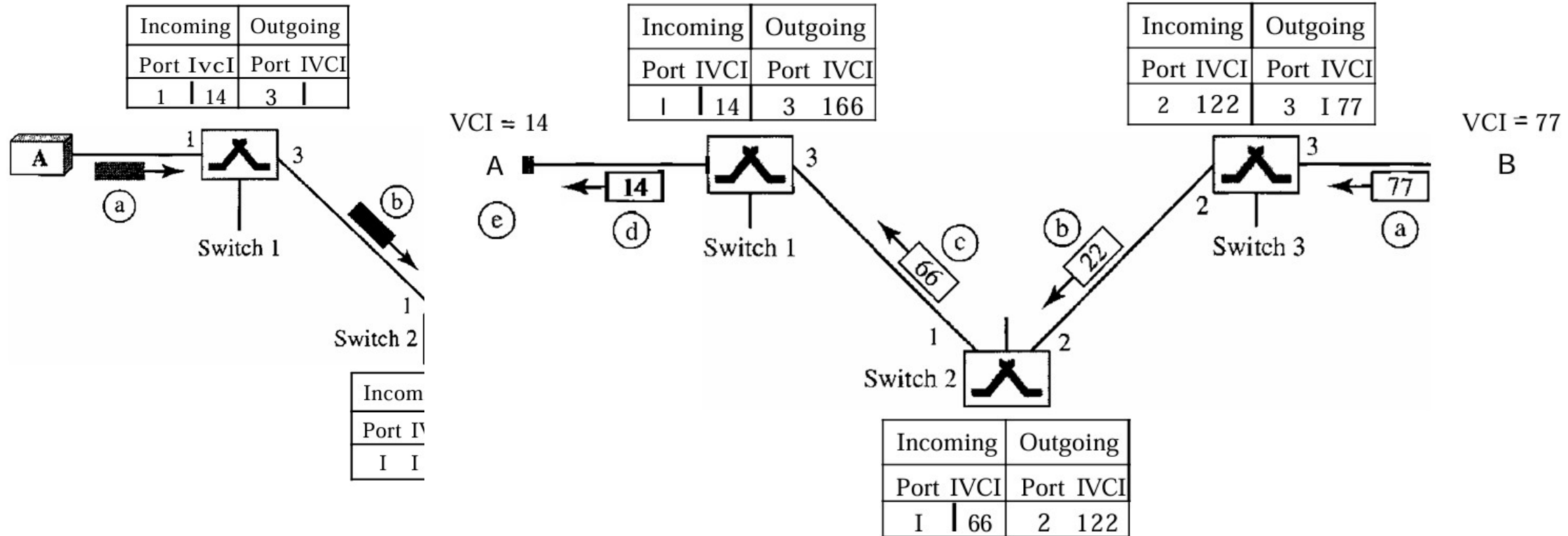
- During the setup phase, the source and destination use their global addresses to create entries in the switches' tables.
- Data transfer takes place after the connection is established.
- In the teardown phase, the source and destination notify the switches to remove the connection entries from their tables.

VCN: Setup Phase

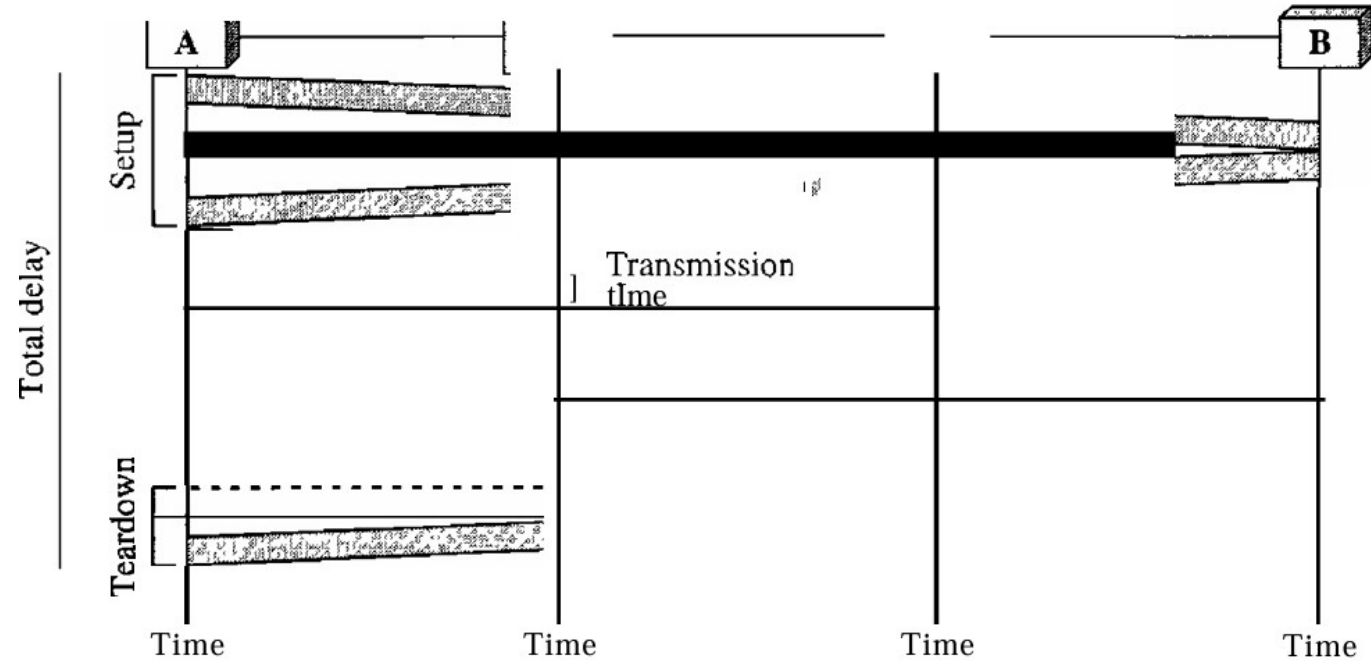
Steps of Setup phase is as follows

- The source first sends a setup request to the destination using its global address.
- Each intermediate switch receives this request and creates a table entry for the connection.
- The switch assigns a local identifier (VCI) for the incoming and outgoing links.
- These table entries together establish a fixed path from source to destination.
- Once the destination accepts the request, a confirmation is sent back, and the connection is ready for data transfer.

VCN Setup: Request



VCN : Delay



$$\text{Total delay} = 3T + 3\tau + \text{setup delay} + \text{teardown delay}$$